The ASIAir, on one page

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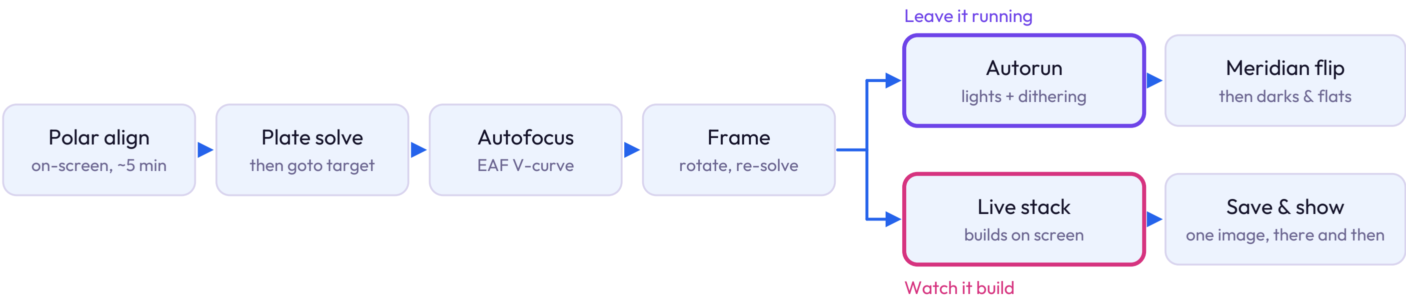
Fragments of the Universe
bhapstar.com

What it is, in one paragraph

A small Linux computer that sits on the telescope. The mount, cameras, focuser and dew heater all plug into it, one 12 volt lead runs to the ground, and you drive the whole night from a phone or tablet over Wi-Fi. Plate solving, autofocus, guiding and the imaging plan all run on the box itself, so the frames keep being written even if the phone goes flat or wanders out of range.

A NIGHT, IN THE ORDER IT HAPPENS



Only the first four stages need you awake. After framing, the night goes one of two ways. Set an Autorun and sleep through it: the box handles the meridian flip on its own, then re-solves and re-centres so the framing survives it. Or switch to Live and let the frames stack on screen as they arrive, which is the mode to use when there are people standing next to you waiting to see something.

WHAT PLUGS INTO WHAT

Mount	goto, tracking, meridian flip
Main camera	exposures, cooling, storage
Guide camera	guiding and dithering
Electronic focuser	autofocus by V-curve
Dew heater	switched 12 V, warm all night
One 12 V input	the only lead running to the ground
Phone or tablet	Wi-Fi, no cable needed

The short cables live on the telescope, which is where most lost sessions come from: a USB lead working loose in the cold, a hub dropping a device, a laptop lid closing. On this rig the controller, main sensor and guide sensor are all inside the ASI585MC Air, so three of those boxes collapse into one.

SETTINGS WORTH GETTING RIGHT

Wi-Fi mode	Station mode	Joins the box to your house network so the phone keeps both
Autofocus	Repeat hourly, or on a temperature drop	A refractor moves focus measurably as the tube cools
Dithering	On, always	Fixed sensor noise lands somewhere new each frame and averages away
Guiding	Below half your image scale	In arcseconds. Keep under that and the stars stay round
Power	One supply, generously sized	Undersized ones fail while cooling and slewing at once, and it looks like a software fault
Storage	Copy off after every night	Short subframes fill the card faster than expected

BEFORE YOUR FIRST NIGHT

Use station mode

Out of the box it broadcasts its own network, so your phone has to leave your home Wi-Fi to reach it. Joining it to the house network keeps both, and lets you check a run from indoors.

Give it real power

A single 12 V feed runs the controller, a cooled camera, the mount and the dew heaters. Undersized supplies cause failures that look like software faults.

Flats before teardown

The moment the camera rotates in the focuser, that night's flats stop matching. Two minutes at the end of the session saves a re-shoot.

It is a closed system

It expects ZWO cameras and a defined list of mounts, not anything that speaks ASCOM. Less choice, and far fewer things to debug in the dark.

WHEN IT GOES WRONG AT THREE IN THE MORNING

What you see	Usually	What to do
Plate solving keeps failing	Too short an exposure, or focus badly out	The solver needs stars it can measure. Get roughly focused first, then lengthen the solve exposure
Autofocus comes back wrong	Thin cloud, or too few stars in the frame	Slew to a richer star field, run it there, then go back to the target
Guiding graph wanders	Polar alignment, or something pulling on a cable	Check nothing is snagged as the mount turns, then re-run polar align. Aim under half your image scale
Cooling or slewing drops out	One 12 V supply doing too much at once	It looks like a software fault and is not. Give it headroom, or split the load
Frames stack badly the next day	Dithering off, or flats taken after teardown	Neither can be fixed later. Leave dithering on, and shoot flats before anything is loosened

The full guide, with examples

Screenshots from a live session, how polar alignment works without seeing Polaris, what plate solving actually does, and where a mini PC running NINA makes more sense than a closed box.

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